

Agricultural Benchmarks differ sharply in scope, labels, and tasks, and current evaluation practice often prevents fair comparison.

The evidence splits into three linked questions: how **PlantVillage**, **iNaturalist**, and **PlantCLEF** represent different benchmark philosophies, which **protocol inconsistencies** make published results hard to compare, and whether **systematic dataset pathologies** have already been documented.

Benchmark Differences

PlantVillage is a **plant disease classification** benchmark built around single cropped leaves photographed under controlled conditions, while iNaturalist and PlantCLEF are **fine-grained plant species recognition** benchmarks collected closer to natural observation settings (Hughes & Salathé, 2015; Picek et al., 2022). PlantVillage contains about **54,309 images**, 14 crop species, and 38 class labels formed as crop–disease or healthy combinations (Hughes & Salathé, 2015; Noyan, 2022). By contrast, iNaturalist 2018 in the plant-recognition literature has **2,917 classes**, 118,800 training images, and 8,751 test images, while PlantCLEF 2017 has **10,000 species** and 320,544 trusted-label training images plus noisy web data at much larger scale (Picek et al., 2022).

PlantVillage’s labels are comparatively **narrow and expert-curated**: experts diagnosed disease states, and only expertly identified leaves were included (Hughes & Salathé, 2015). PlantCLEF 2017 also distinguishes between **trusted labels** and large volumes of **noisy crawled web images**, so its benchmark explicitly mixes clean and noisy supervision (Picek et al., 2022). PlantCLEF 2016 further adds **plant-organ labels** such as flower, leaf, fruit, stem, branch, and whole plant, which changes what the model must learn compared with pure leaf-disease datasets (Picek et al., 2022).

The task definition differs even more than the scale. PlantVillage is mostly treated as **single-image, single-label disease classification** on one whole leaf per image (Dong et al., 2022; Albattah et al., 2021). PlantCLEF since 2017 defines the task at the level of a **plant observation** consisting of one or more images of the same specimen, with the goal of predicting the species (Picek et al., 2022). PlantCLEF 2024 and 2025 pivot again to **multi-label species identification in vegetation plot images**, where the model must predict all species present in a plot rather than a single species or a single disease (Goëau et al., 2025; Filho et al., 2025).

Benchmark Comparison

Benchmark	Scope	Label Quality	Task Definition
PlantVillage	54k controlled crop-leaf images, 14 crops, 38 classes (Hughes & Salathé, 2015)	Expert disease labels, but narrow capture protocol (Hughes & Salathé, 2015)	Single-label leaf disease classification (Dong et al., 2022)
iNaturalist	Thousands of plant species, large train/test splits in naturalist imagery (Picek et al., 2022)	Broad community-scale species labels, used for large-scale recognition benchmarks (Picek et al., 2022)	Fine-grained species recognition (Picek et al., 2022)
PlantCLEF 2017	10,000 species, 320k trusted images plus noisy web data (Picek et al., 2022)	Explicit mixture of trusted and noisy labels (Picek et al., 2022)	Observation-level species prediction from one or more images (Picek et al., 2022)

Benchmark	Scope	Label Quality	Task Definition
PlantCLEF 2024–2025	Plot-scale ecological imagery, >800 species in test, 1.7M individual training images (Goëau et al., 2025)	Expert-annotated multi-label plot test sets (Goëau et al., 2025)	Weakly labeled multi-label species prediction per plot (Goëau et al., 2025; Filho et al., 2025)

FIGURE 1 Scope, labels, and tasks across major benchmarks

A practical consequence is that these benchmarks are **not interchangeable**. PlantVillage tests performance in a clean, constrained phenotyping setup, whereas PlantCLEF and iNaturalist test large-vocabulary fine-grained recognition under more natural observational variability (Huan et al., 2025; Picek et al., 2022). That is why results on one benchmark rarely say much about the others without explicit transfer tests.

Protocol Inconsistencies

Fair comparison is often impossible because papers use **different splits, different class taxonomies, different preprocessing, and different evaluation metrics** (Hu et al., 2026; Chai et al., 2025). Several reviews explicitly say the field remains hard to compare because studies evaluate on **varied datasets** and heterogeneous settings rather than shared protocols (Yao et al., 2023; Hu et al., 2026).

One major inconsistency is the **data split itself**. PlantDoc alone appears in at least two incompatible versions: the original split with 2,360 training and 238 test samples, and a later reshuffled 70/10/20 split called PlantDoc-0.2 (Yao et al., 2023). PlantVillage studies also use different train/validation/test ratios, including 80/10/10, 90/7/3, and sometimes only a random 80/20 split, so headline accuracies are not directly comparable even before architecture differences are considered (Bruno et al., 2022; Noyan, 2022).

Another inconsistency is **leakage control**. Many plant datasets are randomly split under an i.i.d. assumption, which tends to overestimate real deployment performance (Xu et al., 2024). More rigorous work now uses **parent-image-aware** or leakage-safe protocols because crops, leaves, or augmented variants from the same source image can otherwise appear across train and test (Xiang et al., 2026; K et al., 2026). Reviews in grape and general plant disease detection likewise report that **random image splits** usually overstate performance compared with plant-grouped, vineyard-grouped, or external-test evaluation (Ghorpade et al., 2025).

A third inconsistency is that authors evaluate **different tasks under the same headline problem name**. Some papers report whole-image disease classification, others crop–disease pair classification, multi-task species-plus-disease prediction, object detection, or segmentation, each with different label structures and different failure modes (Dong et al., 2022; Moupojou et al., 2023; Wei et al., 2026). Comparing a 99% single-leaf classifier on PlantVillage to a field object detector or plot-level multi-label system is methodologically invalid even if both are called “plant disease recognition” (Moupojou et al., 2023; Goëau et al., 2025).

Common Inconsistencies

Inconsistency	What Varies	Why It Breaks Comparisons
Data splits	80/20, 80/10/10, 90/7/3, reshuffled benchmark variants (Yao et al., 2023; Bruno et al., 2022)	Test difficulty changes materially

Inconsistency	What Varies	Why It Breaks Comparisons
Leakage control	Random image split vs parent-image-aware or grouped split (Xiang et al., 2026; Xu et al., 2024; K et al., 2026)	Inflates reported accuracy
Task definition	Classification, detection, segmentation, multi-task, multi-label (Dong et al., 2022; Goëau et al., 2025; Wei et al., 2026)	Metrics and labels are not equivalent
Taxonomy	Different class sets and crop–disease naming schemes (Hu et al., 2026)	Same “accuracy” may refer to different problems
Metrics	Accuracy only vs macro F1, AUROC, calibration, selective risk (Kondaveeti & Simhadri, 2025; Ghorpade et al., 2025; Xiang et al., 2026)	Different metrics reward different behavior

FIGURE 2 Main protocol inconsistencies across agricultural AI papers

Metric choice is itself a major source of mismatch. Many papers still emphasize **accuracy**, while newer work argues that this misses class imbalance, calibration, OOD behavior, and whether the model used biologically relevant evidence (Saleem et al., 2020; Kondaveeti & Simhadri, 2025; Xiang et al., 2026).

Documented Data Pathologies

Yes, systematic data pathologies have already been identified and quantified in public agricultural datasets. The clearest example is **PlantVillage bias**, where a classifier trained on only **8 background pixels** achieved 49.0% accuracy, far above the 2.6% random baseline, showing that label-correlated noise is present and exploitable (Noyan, 2022). That study attributes the problem mainly to **capture bias**, not classic semantic background bias, and shows that removing the background does not eliminate it (Noyan, 2022).

Other papers document additional pathologies. PlantVillage has strong **class imbalance** and **image redundancy**; tomato-related classes alone make up 43.4% of the images, and many within-class images vary very little (Salman et al., 2025). Field-focused reviews add that many public datasets are **not representative** of real agricultural variability, so even large sample counts can still miss the true range of deployment conditions (Barbedo, 2021).

Annotation pathology is also well documented. PlantDoc is repeatedly criticized for **poor-quality web images**, inclusion of some **laboratory-style images**, and likely **mislabels** because annotations were made without plant pathologists (Moupojou et al., 2023; Moupojou et al., 2024). Broader annotation studies identify multiple inconsistency types in plant disease detection and argue that label strategy itself changes measured model performance (Dong et al., 2022).

Data Pathologies

Pathology	Evidence	Example
Capture bias	Background-only signals predict labels (Noyan, 2022)	PlantVillage 8px gets 49.0% accuracy

Pathology	Evidence	Example
Non-representative acquisition	Controlled images miss field variability (Xiang et al., 2026; Barbedo, 2021)	Lab-to-field collapse
Annotation noise	Labels can be wrong or strategy-dependent (Moupojou et al., 2023; Dong et al., 2022; Xu et al., 2024)	PlantDoc and expert disagreement
Class imbalance	Dominant classes bias learning (Ojo & Zahid, 2023; Salman et al., 2025)	Tomato overrepresentation in PlantVillage
Mixed-domain contamination	Lab and field images coexist in one dataset (Moupojou et al., 2023; Moupojou et al., 2024)	PlantDoc partly contains lab images

FIGURE 3 Systematic pathologies documented in public agricultural datasets

These pathologies are not merely theoretical because they appear in external evaluations. Models trained on PlantVillage often collapse on PlantDoc or other field datasets, with reported drops from around 99% to 31%, from 92.67% to 54.41%, or 20–50 percentage points more generally (Noyan, 2022; Fenu & Mallocci, 2022; Xiang et al., 2026). More recent reliability work adds that these failures are often **overconfident**, not just inaccurate (Xiang et al., 2026).

Evidence Coverage Across Benchmarks

Benchmark Family	Scope	Label quality	Task clarity	Protocol standardization
PlantVillage	10	8	9	4
PlantDoc and field disease sets	8	7	7	3
iNaturalist / PlantCLEF species benchmarks	7	6	8	6
Segmentation / detection field benchmarks	5	5	6	2

FIGURE 4 Evidence coverage across benchmark families and evaluation issues

The best-covered area is the critique of **PlantVillage-style controlled benchmarking**, especially its gap to field deployment (Xu et al., 2024; B et al., 2026; Jelali, 2024). The biggest persistent gap is **protocol standardization across field tasks**, where recent papers still argue that benchmark datasets and evaluation frameworks are insufficient for unseen crops, domain shift, and multi-crop disease recognition (Chai et al., 2025; Caroppo et al., 2026).

What This Means

The most important shift is that the field is moving from asking **which model gets the highest random-split accuracy on PlantVillage** to asking **which benchmark and protocol actually measure deployable agricultural vision** (Xiang et al., 2026; Chai et al., 2025; Xu et al., 2024). PlantVillage remains useful for controlled algorithmic comparison, but it is a poor proxy for ecological species recognition or field disease diagnosis, which are closer to iNaturalist-, PlantCLEF-, and real-field dataset regimes (Yao et al., 2023; Huan et al., 2025).

The single biggest open question is how to build **shared, leakage-safe, task-aligned benchmarks** that combine expert labels, realistic field variability, explicit taxonomies, and standardized external-test protocols across crops, diseases, and sensing conditions (Xu et al., 2024; Chai et al., 2025). On your three benchmark questions, the answer is therefore yes on all fronts: the benchmarks differ fundamentally, current protocol heterogeneity often blocks fair comparison, and systematic dataset pathologies have already been clearly documented.

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